```
entry:
%0 = icmp eq i32 %num_args, 2, !dbg !15
 br i1 %0, label %assert_end, label %assert_fail, !dbg !15, !prof !16
T | F
```

```
assert_end:
%.not = icmp eq ptr %args, null, !dbg !15
 br i1 %.not, label %assert_fail1, label %assert_end2, !dbg !15, !prof !20
T | F
```

```
assert_end2:
%var_A.type_index = load i32, ptr %args, align 4, !dbg !15
 %var_A.type_index.fr = freeze i32 %var_A.type_index, !dbg !15
 %3 = icmp sgt i32 %var_A.type_index.fr, 63, !dbg !15
 br i1 %3, label %assert_end4, label %switch.early.test, !dbg !15
T | F
```

```
switch.early.test:
switch i32 %var_A.type_index.fr, label %assert_fail3 [
 i32 7, label %assert_end4
 i32 4, label %assert_end4
 i32 0, label %assert_end4
 ], !dbg !15
def | 7 | 4 | 0
```

```
assert_fail:
%1 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %1(ptr nonnull @.str.1, ptr nonnull @.str), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end4:
%5 = getelementptr inbounds nuw i8, ptr %args, i64 16, !dbg !15
 %var_PadInput.type_index = load i32, ptr %5, align 4, !dbg !15
 %var_PadInput.type_index.fr = freeze i32 %var_PadInput.type_index, !dbg !15
 %6 = icmp sgt i32 %var_PadInput.type_index.fr, 63, !dbg !15
 br i1 %6, label %assert_end6, label %switch.early.test43, !dbg !15
T | F
```

```
assert_fail1:
%2 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %2(ptr nonnull @.str.1, ptr nonnull @.str.2), !dbg !15
 br label %common.ret, !dbg !15
```

```
switch.early.test43:
switch i32 %var_PadInput.type_index.fr, label %assert_fail5 [
 i32 7, label %assert_end6
 i32 4, label %assert_end6
 i32 0, label %assert_end6
 ], !dbg !15
def | 7 | 4 | 0
```

```
assert_end6:
%8 = getelementptr inbounds nuw i8, ptr %args, i64 8, !dbg !15
 %9 = load ptr, ptr %8, align 8, !dbg !15
 %10 = icmp eq i32 %var_A.type_index.fr, 72, !dbg !15
 %var_A.idx = select i1 %10, i64 16, i64 0, !dbg !15
 %var_A = getelementptr inbounds nuw i8, ptr %9, i64 %var_A.idx, !dbg !15
 %11 = getelementptr inbounds nuw i8, ptr %args, i64 24, !dbg !15
 %12 = load ptr, ptr %11, align 8, !dbg !15
 %13 = icmp eq i32 %var_PadInput.type_index.fr, 72, !dbg !15
 %var_PadInput.idx = select i1 %13, i64 16, i64 0, !dbg !15
 %var_PadInput = getelementptr inbounds nuw i8, ptr %12, i64
... %var_PadInput.idx, !dbg !15
 %.not45 = icmp eq ptr %9, null, !dbg !15
 br i1 %.not45, label %assert_fail7, label %assert_end8, !dbg !15, !prof !20
T | F
```

```
assert_fail3:
%4 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %4(ptr nonnull @.str.1, ptr nonnull @.str.3), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_fail5:
%7 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %7(ptr nonnull @.str.1, ptr nonnull @.str.4), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_fail7:
%14 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %14(ptr nonnull @.str.1, ptr nonnull @.str.5), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end8:
%15 = getelementptr inbounds nuw i8, ptr %var_A, i64 16, !dbg !15
 %16 = load i32, ptr %15, align 4, !dbg !15
 %17 = icmp eq i32 %16, 2, !dbg !15
 br i1 %17, label %assert_end10, label %assert_fail9, !dbg !15, !prof !16
T | F
```

```
assert_end10:
%19 = getelementptr inbounds nuw i8, ptr %var_A, i64 24, !dbg !15
 %pad.var_A.shape = load ptr, ptr %19, align 8, !dbg !15
 %20 = load i64, ptr %pad.var_A.shape, align 8, !dbg !15, !tbaa !28
 %M = trunc i64 %20 to i32, !dbg !15
 %21 = getelementptr inbounds nuw i8, ptr %pad.var_A.shape, i64 8, !dbg !15
 %22 = load i64, ptr %21, align 8, !dbg !15, !tbaa !39
 %N = trunc i64 %22 to i32, !dbg !15
 %23 = getelementptr inbounds nuw i8, ptr %var_A, i64 32, !dbg !15
 %pad.var_A.strides = load ptr, ptr %23, align 8, !dbg !15
 %24 = getelementptr inbounds nuw i8, ptr %var_A, i64 12, !dbg !15
 %dev_id = load i32, ptr %24, align 4, !dbg !15
 %A = load ptr, ptr %var_A, align 8, !dbg !15
 call void @llvm.assume(i1 true) [ "align"(ptr %A, i64 64) ], !dbg !15
 %.not46 = icmp eq ptr %12, null, !dbg !15
 br i1 %.not46, label %assert_fail11, label %assert_end12, !dbg !15, !prof !20
T | F
```

```
assert_fail9:
%18 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %18(ptr nonnull @.str.1, ptr nonnull @.str.6), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_fail11:
%25 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %25(ptr nonnull @.str.1, ptr nonnull @.str.7), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end12:
%26 = getelementptr inbounds nuw i8, ptr %var_PadInput, i64 16, !dbg !15
 %27 = load i32, ptr %26, align 4, !dbg !15
 %28 = icmp eq i32 %27, 2, !dbg !15
 br i1 %28, label %assert_end14, label %assert_fail13, !dbg !15, !prof !16
T | F
```

```
assert_end14:
%30 = getelementptr inbounds nuw i8, ptr %var_PadInput, i64 24, !dbg !15
 %pad.var_PadInput.shape = load ptr, ptr %30, align 8, !dbg !15
 %31 = getelementptr inbounds nuw i8, ptr %var_PadInput, i64 32, !dbg !15
 %pad.var_PadInput.strides = load ptr, ptr %31, align 8, !dbg !15
 %PadInput = load ptr, ptr %var_PadInput, align 8, !dbg !15
 call void @llvm.assume(i1 true) [ "align"(ptr %PadInput, i64 64) ], !dbg !15
 %32 = getelementptr inbounds nuw i8, ptr %var_A, i64 22, !dbg !15
 %33 = load i16, ptr %32, align 2, !dbg !15
 %34 = icmp eq i16 %33, 1, !dbg !15
 %35 = getelementptr inbounds nuw i8, ptr %var_A, i64 21, !dbg !15
 %36 = load i8, ptr %35, align 1, !dbg !15
 %37 = icmp eq i8 %36, 32, !dbg !15
 %38 = getelementptr inbounds nuw i8, ptr %var_A, i64 20, !dbg !15
 %39 = load i8, ptr %38, align 1, !dbg !15
 %40 = icmp eq i8 %39, 2, !dbg !15
 %41 = and i1 %37, %40, !dbg !15
 %42 = and i1 %34, %41, !dbg !15
 br i1 %42, label %assert_end16, label %assert_fail15, !dbg !15, !prof !16
T | F
```

```
assert_fail13:
%29 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %29(ptr nonnull @.str.1, ptr nonnull @.str.8), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end16:
%.not47 = icmp eq ptr %pad.var_A.strides, null, !dbg !15
 br i1 %.not47, label %if_end, label %if_then, !dbg !15, !prof !20
T | F
```

```
assert_fail15:
%43 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %43(ptr nonnull @.str.1, ptr nonnull @.str.9), !dbg !15
 br label %common.ret, !dbg !15
```

```
if_then:
%44 = load i64, ptr %pad.var_A.strides, align 8, !dbg !15, !tbaa !50
 %45 = trunc i64 %44 to i32, !dbg !15
 %46 = icmp eq i32 %N, %45, !dbg !15
 %47 = icmp eq i32 %M, 1, !dbg !15
 %48 = or i1 %47, %46, !dbg !15
 %49 = getelementptr inbounds nuw i8, ptr %pad.var_A.strides, i64 8, !dbg !15
 %50 = load i64, ptr %49, align 8, !dbg !15, !tbaa !60
 %51 = and i64 %50, 4294967295, !dbg !15
 %52 = icmp eq i64 %51, 1, !dbg !15
 %53 = icmp eq i32 %N, 1, !dbg !15
 %54 = or i1 %53, %52, !dbg !15
 %55 = and i1 %48, %54, !dbg !15
 br i1 %55, label %if_end, label %assert_fail17, !dbg !15, !prof !16
T | F
```

```
if_end:
%56 = getelementptr inbounds nuw i8, ptr %var_A, i64 40, !dbg !15
 %57 = load i64, ptr %56, align 8, !dbg !15
 %58 = icmp eq i64 %57, 0, !dbg !15
 br i1 %58, label %assert_end20, label %assert_fail19, !dbg !15, !prof !16
T | F
```

```
assert_fail17:
%59 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %59(ptr nonnull @.str.1, ptr nonnull @.str.10), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end20:
%61 = getelementptr inbounds nuw i8, ptr %var_A, i64 8, !dbg !15
 %62 = load i32, ptr %61, align 4, !dbg !15
 %63 = icmp eq i32 %62, 1, !dbg !15
 br i1 %63, label %assert_end22, label %assert_fail21, !dbg !15, !prof !16
T | F
```

```
assert_fail19:
%60 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %60(ptr nonnull @.str.1, ptr nonnull @.str.11), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end22:
%65 = icmp ne ptr %A, null, !dbg !15
 %66 = mul nsw i32 %N, %M, !dbg !15
 %67 = icmp eq i32 %66, 0, !dbg !15
 %68 = or i1 %67, %65, !dbg !15
 br i1 %68, label %assert_end24, label %assert_fail23, !dbg !15, !prof !16
T | F
```

```
assert_fail21:
%64 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %64(ptr nonnull @.str.1, ptr nonnull @.str.12), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end24:
%70 = getelementptr inbounds nuw i8, ptr %var_PadInput, i64 22, !dbg !15
 %71 = load i16, ptr %70, align 2, !dbg !15
 %72 = icmp eq i16 %71, 1, !dbg !15
 %73 = getelementptr inbounds nuw i8, ptr %var_PadInput, i64 21, !dbg !15
 %74 = load i8, ptr %73, align 1, !dbg !15
 %75 = icmp eq i8 %74, 32, !dbg !15
 %76 = getelementptr inbounds nuw i8, ptr %var_PadInput, i64 20, !dbg !15
 %77 = load i8, ptr %76, align 1, !dbg !15
 %78 = icmp eq i8 %77, 2, !dbg !15
 %79 = and i1 %75, %78, !dbg !15
 %80 = and i1 %72, %79, !dbg !15
 br i1 %80, label %assert_end26, label %assert_fail25, !dbg !15, !prof !16
T | F
```

```
assert_fail23:
%69 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %69(ptr nonnull @.str.1, ptr nonnull @.str.13), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end26:
%82 = add nsw i32 %M, 4, !dbg !15
 %83 = load i64, ptr %pad.var_PadInput.shape, align 8, !dbg !15, !tbaa !62
 %84 = trunc i64 %83 to i32, !dbg !15
 %85 = icmp eq i32 %82, %84, !dbg !15
 br i1 %85, label %assert_end28, label %assert_fail27, !dbg !15, !prof !16
T | F
```

```
assert_fail25:
%81 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %81(ptr nonnull @.str.1, ptr nonnull @.str.14), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end28:
%87 = add nsw i32 %N, 4, !dbg !15
 %88 = getelementptr inbounds nuw i8, ptr %pad.var_PadInput.shape, i64 8,
... !dbg !15
 %89 = load i64, ptr %88, align 8, !dbg !15, !tbaa !72
 %90 = trunc i64 %89 to i32, !dbg !15
 %91 = icmp eq i32 %87, %90, !dbg !15
 br i1 %91, label %assert_end30, label %assert_fail29, !dbg !15, !prof !16
T | F
```

```
assert_fail27:
%86 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %86(ptr nonnull @.str.1, ptr nonnull @.str.15), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end30:
%.not48 = icmp eq ptr %pad.var_PadInput.strides, null, !dbg !15
 br i1 %.not48, label %if_end32, label %if_then31, !dbg !15, !prof !20
T | F
```

```
assert_fail29:
%92 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %92(ptr nonnull @.str.1, ptr nonnull @.str.16), !dbg !15
 br label %common.ret, !dbg !15
```

```
if_then31:
%93 = load i64, ptr %pad.var_PadInput.strides, align 8, !dbg !15, !tbaa !74
 %94 = trunc i64 %93 to i32, !dbg !15
 %95 = icmp eq i32 %87, %94, !dbg !15
 %96 = icmp eq i32 %82, 1, !dbg !15
 %97 = or i1 %96, %95, !dbg !15
 %98 = getelementptr inbounds nuw i8, ptr %pad.var_PadInput.strides, i64 8,
... !dbg !15
 %99 = load i64, ptr %98, align 8, !dbg !15, !tbaa !84
 %100 = and i64 %99, 4294967295, !dbg !15
 %101 = icmp eq i64 %100, 1, !dbg !15
 %102 = icmp eq i32 %87, 1, !dbg !15
 %103 = or i1 %102, %101, !dbg !15
 %104 = and i1 %97, %103, !dbg !15
 br i1 %104, label %if_end32, label %assert_fail33, !dbg !15, !prof !16
T | F
```

```
if_end32:
%105 = getelementptr inbounds nuw i8, ptr %var_PadInput, i64 40, !dbg !15
 %106 = load i64, ptr %105, align 8, !dbg !15
 %107 = icmp eq i64 %106, 0, !dbg !15
 br i1 %107, label %assert_end36, label %assert_fail35, !dbg !15, !prof !16
T | F
```

```
assert_fail33:
%108 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %108(ptr nonnull @.str.1, ptr nonnull @.str.17), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end36:
%110 = getelementptr inbounds nuw i8, ptr %var_PadInput, i64 8, !dbg !15
 %111 = load i32, ptr %110, align 4, !dbg !15
 %112 = icmp eq i32 %111, 1, !dbg !15
 br i1 %112, label %assert_end38, label %assert_fail37, !dbg !15, !prof !16
T | F
```

```
assert_fail35:
%109 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %109(ptr nonnull @.str.1, ptr nonnull @.str.18), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end38:
%114 = getelementptr inbounds nuw i8, ptr %var_PadInput, i64 12, !dbg !15
 %115 = load i32, ptr %114, align 4, !dbg !15
 %116 = icmp eq i32 %dev_id, %115, !dbg !15
 br i1 %116, label %assert_end40, label %assert_fail39, !dbg !15, !prof !16
T | F
```

```
assert_fail37:
%113 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %113(ptr nonnull @.str.1, ptr nonnull @.str.19), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end40:
%118 = icmp ne ptr %PadInput, null, !dbg !15
 %119 = mul nsw i32 %87, %82, !dbg !15
 %120 = icmp eq i32 %119, 0, !dbg !15
 %121 = or i1 %120, %118, !dbg !15
 br i1 %121, label %assert_end42, label %assert_fail41, !dbg !15, !prof !16
T | F
```

```
assert_fail39:
%117 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %117(ptr nonnull @.str.1, ptr nonnull @.str.20), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_end42:
tail call fastcc void @pad_compute_(i32 %M, i32 %N, ptr %PadInput, ptr
... %A), !dbg !15
 br label %common.ret, !dbg !15
```

```
assert_fail41:
%122 = load ptr, ptr @__TVMFFIErrorSetRaisedFromCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %122(ptr nonnull @.str.1, ptr nonnull @.str.21), !dbg !15
 br label %common.ret, !dbg !15
```

```
common.ret:
%common.ret.op = phi i32 [ -1, %assert_fail ], [ -1, %assert_fail1 ], [
... -1, %assert_fail3 ], [ -1, %assert_fail5 ], [ -1, %assert_fail7 ], [ -1,
... %assert_fail9 ], [ -1, %assert_fail11 ], [ -1, %assert_fail13 ], [ -1,
... %assert_fail15 ], [ -1, %assert_fail17 ], [ -1, %assert_fail19 ], [ -1,
... %assert_fail21 ], [ -1, %assert_fail23 ], [ -1, %assert_fail25 ], [ -1,
... %assert_fail27 ], [ -1, %assert_fail29 ], [ -1, %assert_fail33 ], [ -1,
... %assert_fail35 ], [ -1, %assert_fail37 ], [ -1, %assert_fail39 ], [ -1,
... %assert_fail41 ], [ 0, %assert_end42 ]
 ret i32 %common.ret.op, !dbg !15
```

CFG for 'pad' function